Adaptive Portfolio Positioning under Market Volatility

A Forward-Looking Regime-Switching Approach
Using the VIX Term Structure

Liu Chun Hin

The Hong Kong Polytechnic University

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Abstract

Large maximum drawdowns permanently impair the terminal wealth of long-horizon investors, even when subsequent average returns recover. We present a transparent, implementable **dynamic risk-overlay** framework that combines daily regime detection with weekly-scale execution constraints. A two-state Markov-switching GJR-GARCH model with time-varying transition probabilities driven by lagged log(VIX) and the VIX term-structure inversion generates forward-looking stress probabilities. These probabilities are mapped into continuous equity weights through a soft threshold rule, subject to a five-trading-day minimum holding period and stylized tax considerations.

Under strict expanding-window out-of-sample evaluation on SPY (2010–2026), the overlay reduces annualized volatility by approximately **25 percent** and maximum drawdown by roughly **six percentage points** relative to buy-and-hold, while keeping turnover modest and preserving a comparable Sharpe ratio. A three-factor extension that adds the lagged change in log(VIX) does not improve risk metrics or coverage of major stress episodes. The binding constraint on out-of-sample risk reduction is frequently the mapping from probability to position size rather than the precise specification of the transition equation. The contribution is a practical downside-protection rule that limits large drawdowns without claiming superior risk-adjusted returns.

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1. Introduction

Large maximum drawdowns rank among the most damaging events for long-horizon investors. They interrupt compounding, induce poorly timed exits, and can permanently reduce terminal wealth even when later recoveries restore average returns. The high-volatility episodes of 2020–2026 have underscored this vulnerability and renewed interest in systematic rules that reduce equity exposure during stress periods (Moreira and Muir, 2017, 2019; Barroso and Santa-Clara, 2015).

Volatility-managed and regime-based strategies offer a natural response. Scaling positions inversely with realized or implied volatility has been shown to improve risk-adjusted outcomes (Moreira and Muir, 2017; Bongaerts et al., 2020; Cederburg et al., 2020). Markov-switching models with time-varying transition probabilities provide a flexible statistical framework for detecting shifts between normal and high-volatility regimes (Hamilton, 1989; Filardo, 1994; Guidolin and Timmermann, 2007). The VIX and its term structure have long been viewed as forward-looking fear gauges (Whaley, 2000; Bates, 2012).

Translating these insights into rules that investors can actually implement remains incomplete. Slope-only inversion signals frequently fail during prolonged grind-down markets, most notably in 2022. Unrestricted multi-state Markov models tend to produce unrealistically short regime durations. Existing studies also typically abstract from practical frictions—transaction costs, capital-gains tax considerations, and the need for multi-day position persistence—leaving a gap between statistical detection and feasible execution.

This paper closes that gap. We develop a hybrid two-state time-varying transition probability (TVTP) Markov-switching GJR-GARCH model that incorporates the lagged log(VIX) level and the VIX inversion ratio as transition drivers, subject to economically motivated sign restrictions and asymmetric soft persistence penalties. Regime probabilities are mapped into continuous portfolio weights through a soft threshold rule calibrated to maintain moderate average equity exposure. Execution is further constrained by a five-trading-day minimum holding period and stylized tax drag, yielding a strategy that detects regimes at daily frequency yet trades at weekly scale.

Our primary evaluation relies on strict expanding-window out-of-sample testing. The resulting dynamic risk overlay reduces annualized volatility by approximately 25 percent and maximum drawdown by roughly six percentage points relative to buy-and-hold, while keeping turnover controlled and delivering a Sharpe ratio at least as high as the benchmark. A three-factor extension that adds the lagged change in log(VIX) does not improve either risk metrics or coverage of major stress episodes. Sensitivity analysis shows that the execution-layer mapping from regime probability to position size is often the binding determinant of out-of-sample risk reduction—more so than the precise coefficients of the transition equation.

The contribution is therefore pragmatic: a transparent downside-protection rule that demonstrably limits large drawdowns without claiming superior Sharpe ratios under rigorous out-of-sample conditions. The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data, model, and position engine. Section 4 diagnoses the strengths and failure modes of the inversion signal. Section 5 presents detection diagnostics and expanding-window performance. Section 6 discusses practical implementation considerations for institutional risk overlays and systematic wealth managers. Section 7 concludes.

2. Related Research

The literature on volatility timing is extensive. Moreira and Muir (2017) show that scaling equity exposure inversely with realized variance can produce substantial improvements in Sharpe ratios. Subsequent work has examined robustness, transaction costs, and investor clienteles (Cederburg et al., 2020; Bongaerts et al., 2020; Moreira and Muir, 2019). Related research on momentum crashes and volatility-managed factor portfolios further underscores the practical value of conditioning on risk (Barroso and Santa-Clara, 2015; Daniel and Moskowitz, 2016). These papers establish that volatility management is economically meaningful, yet most implementations rely on simple inverse-volatility rules rather than forward-looking regime detection.

Markov-switching models provide a natural statistical framework for identifying discrete shifts between market states (Hamilton, 1989, 2005). Filardo (1994) extended the framework to time-varying transition probabilities, allowing exogenous covariates to influence the likelihood of regime changes. Guidolin and Timmermann (2007, 2008) apply multivariate regime-switching methods to asset allocation and show that regime awareness can improve out-of-sample portfolio performance. Despite these advances, most implementations remain difficult to translate into practical rules because they produce either excessively frequent switches or require continuous rebalancing.

A second strand of research focuses on the information content of the VIX and the VIX term structure (Whaley, 2000; Bates, 2012; Johnson, 2011). The near-term to medium-term inversion ratio has been interpreted as a forward-looking signal of demand for short-dated protection. While this signal is informative during acute panic episodes, it is less reliable in prolonged grind-down markets where the entire curve shifts upward without sharp inversion.

Practical constraints such as transaction costs and position persistence have received comparatively less attention in the regime-switching literature. Grossman and Laroque (1990) analyze (S, s) rules arising from transaction costs, and practical discussions of position sizing appear in the broader portfolio-management literature (Thorp, 2006). Few studies combine daily statistical detection with multi-day minimum holding periods, soft position mapping, and stylized tax considerations—features that are material for long-horizon investors concerned with large maximum drawdowns.

This paper sits at the intersection of these literatures. We embed the VIX level and inversion ratio inside a two-state TVTP Markov-switching GJR-GARCH model, impose economically motivated sign restrictions and asymmetric persistence penalties, and map filtered probabilities into continuous weights under explicit trading constraints. The resulting hybrid design—daily detection, weekly-scale execution—addresses a practical gap between statistical regime identification and implementable risk management.

3. Data and Methodology

3.1. Data

We use daily data on SPY, the VIX, the three-month VIX (VIX3M), and the short-term Treasury ETF BIL from January 2010 through July 2026. All series are obtained from standard market-data sources. Excess returns are computed relative to the BIL return. The key transition variables are the lagged natural logarithm of the VIX and the lagged inversion ratio $\mathrm{VIX}_t/\mathrm{VIX3M}_t$. A three-factor extension additionally includes the lagged first difference of log(VIX). All predictors are lagged by at least one day to ensure that signals are measurable before the return they are asked to forecast. The final daily sample contains approximately 4,150 observations.

3.2. Regime Model

We estimate a two-state Markov-switching GJR-GARCH(1,1) model with time-varying transition probabilities. Let $s_t \in \{0, 1\}$ denote the latent regime (0 = Normal Growth, 1 = High-Vol / Stress). Within each regime the conditional mean and variance follow

$$r_t = \mu_{s_t} + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad z_t \sim N(0,1), \tag{1}$$

$$\sigma_t^2 = \omega_{s_t} + \big(\alpha_{s_t} + \gamma_{s_t} I_{\{\varepsilon_{t-1}<0\}}\big)\varepsilon_{t-1}^2 + \beta_{s_t}\sigma_{t-1}^2. \tag{2}$$

Dual identification is imposed by ordering regimes so that $\mu_0 > \mu_1$ and $\omega_0 < \omega_1$.

Transition probabilities are driven by a vector of lagged covariates z_{t-1}. For the main two-factor specification,

$$z_{t-1} = \big[\log(\mathrm{VIX}_{t-1}),\ \mathrm{Inversion}_{t-1}\big].$$

The three-factor extension appends $\Delta \log(\mathrm{VIX})_{t-1}$. Entry into the Stress regime is restricted to respond non-negatively to the covariates, while exit coefficients remain unrestricted. Soft asymmetric persistence penalties encourage a high stay probability in the Normal regime (approximately 0.97) and a moderately lower stay probability in Stress (approximately 0.90), together with an unconditional Stress mass target near 0.25. Parameters are estimated by maximum likelihood using multi-start L-BFGS-B.

3.3. Position Engine

Filtered Stress probabilities $P(s_t = 1 \mid \mathcal{F}_t)$ are converted into continuous equity weights by a soft threshold rule:

$$w_t = 1 + (W_{\mathrm{STRESS}} - 1)\cdot \mathrm{clip}\Big(\frac{P(s_t = 1 \mid \mathcal{F}_t)}{\tau},\ 0,\ 1\Big), \tag{3}$$

with frozen parameters $\tau = 0.65$ and $W_{\mathrm{STRESS}} = 0.50$ (hence full exposure in the Normal regime). An (S, s)-style band and a five-trading-day minimum holding period further limit turnover. Transaction costs are charged at a fixed number of basis points per unit of turnover; a stylized short-term /

long-term capital-gains tax is applied when positions are reduced after a gain. The resulting strategy detects regimes daily but effectively trades at a weekly horizon.

Figure 1 illustrates the soft-mapping function. The frozen calibration produces a smooth transition from full exposure to a 50 percent Stress weight once the probability exceeds the threshold. This continuous mapping, combined with the minimum-holding constraint, is the key mechanism that converts short-lived Stress signals into implementable risk reduction.

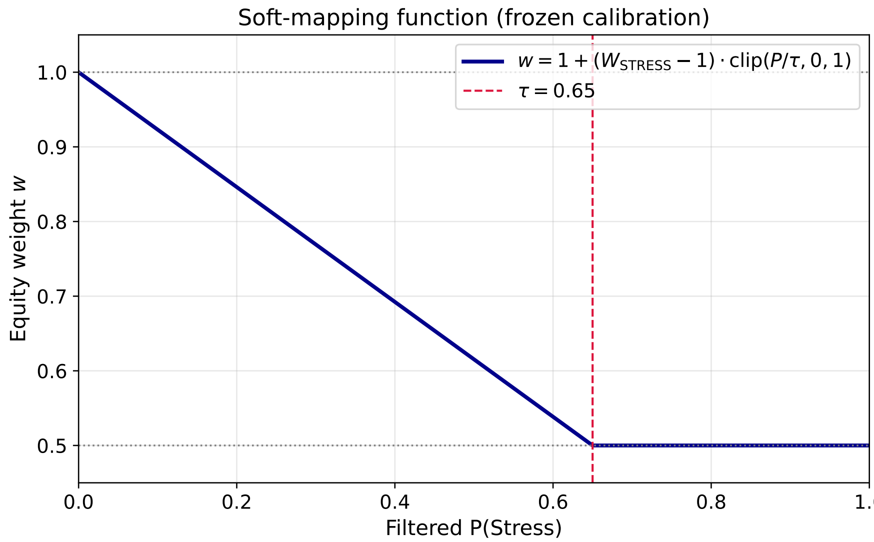

Figure 1: Soft-mapping function from filtered Stress probability to equity weight. The frozen parameters $\tau = 0.65$ and $W_{\mathrm{STRESS}} = 0.50$ generate a gradual reduction in exposure once the probability exceeds the threshold. This design avoids the binary behavior of hard thresholding while keeping average equity weight in the economically sensible range of 0.75–0.85.

3.4. Evaluation Design

All performance claims rest on strict expanding-window out-of-sample evaluation. The model is first estimated on an initial window of at least 1,500 daily observations and then re-estimated approximately every 130 trading days. Portfolio weights are generated only with information available at each re-estimation date. We report annualized return, volatility, maximum drawdown, Sharpe ratio, turnover, and—in a compact companion table—Calmar ratio, 95 percent Conditional Value-at-Risk (CVaR), and downside deviation. Full-sample results are presented solely for illustration; the expanding-window exercise is the primary evidence. An episode table covering the major stress windows of 2011, 2015, 2018, 2020, 2022, and 2024 is used to assess regime-detection quality under a transparent under-detection rule.

To place the strategy in a practitioner context, we also compare it against three simple benchmarks that investors might actually implement: (i) Moreira–Muir style inverse-volatility scaling, (ii) a traditional 200-day moving-average timing rule, and (iii) a naïve VIX-level threshold rule (reduce exposure when VIX exceeds 25).

4. Inversion Diagnosis

The VIX term-structure inversion ratio (near-term VIX divided by the three-month VIX) is an attractive candidate for a forward-looking stress indicator. It is scale-free, observable in real time, and can be lagged without look-ahead bias. In acute panic episodes the ratio rises sharply, correctly signaling elevated demand for short-dated protection. These properties motivate its inclusion as a transition driver.

Nevertheless, the signal possesses well-documented limitations. First, parallel upward shifts of the entire volatility curve can occur without inversion; in such cases the ratio remains near or below unity even while equity prices decline substantially. Second, the distinction between a transitory spike and a sustained high-volatility regime is imperfectly captured by a single static threshold. Third, the ratio is fragile to the precise definition of "near-term" versus "medium-term," and small changes in the maturity pair can alter the timing of signals.

These weaknesses are most visible in prolonged grind-down markets. During the 2022 bear market the inversion ratio exceeded one on only about 7 percent of trading days, yet SPY experienced a maximum drawdown of approximately 25 percent. Similar, though less severe, under-detection occurred in the 2015 China-related episode. In contrast, the acute stress windows of 2011, 2018 (Volmageddon), 2020 (COVID), and August 2024 produced clearer inversion signals and correspondingly higher filtered Stress probabilities.

Figure 2 focuses on the two most informative failure cases. In both episodes the inversion ratio remains subdued for extended periods while equity prices decline. The two-factor model elevates Stress probability relative to an inversion-only benchmark, yet the elevation is still insufficient to clear a transparent under-detection threshold. Adding the lagged change in log(VIX) produces only marginal improvement in 2022 and actually lowers mean Stress probability in 2015.

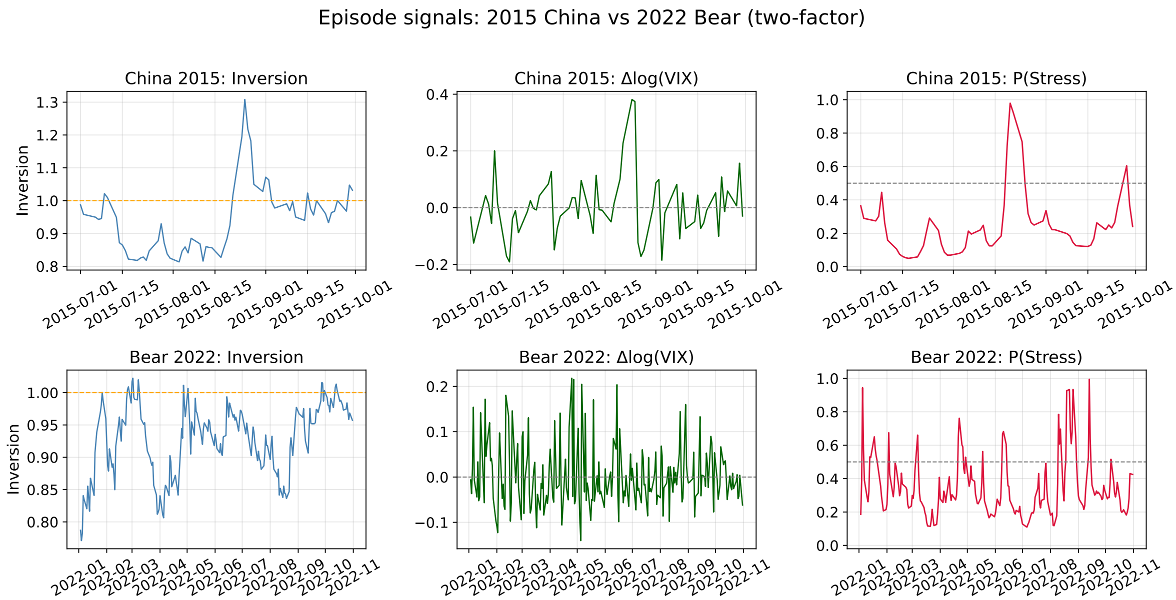

Figure 2: Inversion ratio, lagged change in log(VIX), and filtered Stress probability during the 2015 China episode (upper panels) and the 2022 bear market (lower panels). Shaded regions indicate periods when SPY drawdown exceeds 5 percent. The inversion signal is weak in both grind-down markets; the two-factor and three-factor models only partially compensate.

We therefore treat inversion as a useful but incomplete driver. The two-factor specification that combines the lagged log(VIX) level with the inversion ratio partially mitigates the 2022 failure, yet still falls short of the under-detection rule. The three-factor extension does not resolve the structural limitations and is retained solely for robustness.

5. Empirical Results

5.1. Detection Diagnostics

Table 1 summarizes the key detection-layer statistics for both specifications. Expected duration in the Normal regime is approximately 13 days; expected duration in the Stress regime remains short at roughly 3.5 days. Stationary Stress mass is near 0.21–0.22. These magnitudes are consistent with the soft persistence penalties, yet the short Stress duration constitutes a structural limitation: hard threshold rules rarely trigger de-risking and therefore fail to reduce maximum drawdown relative to buy-and-hold.

Table 1: Detection-Layer Summary

Metric	Two-factor	Three-factor
s_{enter}	[0.42, 0.49]	[0.23, 0.51, 0.66]
E[duration] Normal / Stress (days)	12.6 / 3.5	13.6 / 3.5
Stationary Stress mass	≈ 0.22	≈ 0.21
Smoothed Stress frequency	≈ 0.09	≈ 0.08
Mean P(Stress)	≈ 0.20	≈ 0.19

Figure 3 plots the full-sample filtered Stress probability together with the six major stress windows examined in the episode table. Acute episodes (2011, 2018, 2020, 2024) produce clear elevations in probability. The grind-down episodes (2015, 2022) generate weaker and more intermittent signals, consistent with the diagnosis in Section 4.

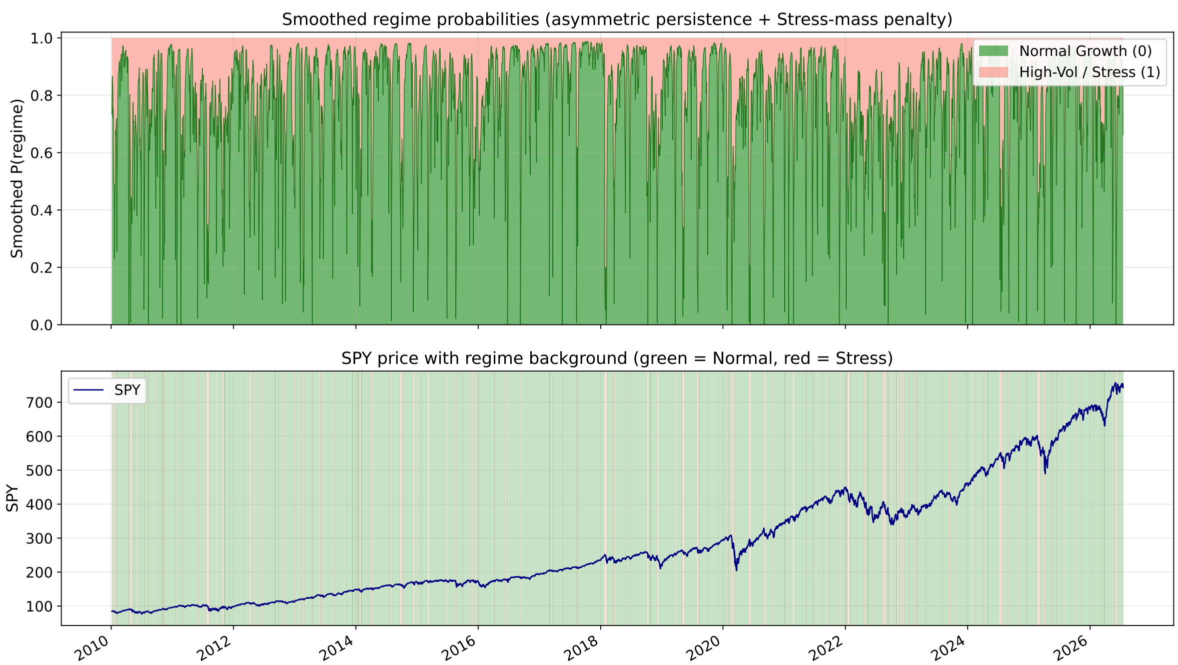

Figure 3: Filtered Stress probability (two-factor model) over the full sample. Shaded vertical bands mark the six major stress windows used in the episode diagnosis. Acute panic episodes generate sharp rises in probability; grind-down episodes produce weaker and more intermittent signals.

Table 2 reports the episode diagnosis under the under-detection rule. The two-factor model flags China 2015 and Bear 2022. The three-factor model clears the 2022 flag only marginally and worsens the 2015 flag. Overall episode coverage is not systematically superior.

Table 2: Episode Diagnosis (Smoothed P(Stress))

Episode	MinDD	Frac Inv>1	2F Mean P	2F FLAG	3F Mean P	3F FLAG
EU 2011	−18.6%	56.8%	**0.341**	ok	0.322	ok
China 2015	−11.9%	23.4%	**0.249**	FLAG	0.225	FLAG
Volmageddon	−10.1%	35.5%	**0.321**	ok	0.297	ok
COVID 2020	−33.7%	68.3%	**0.305**	ok	0.249	ok
Bear 2022	−24.5%	7.2%	0.346	FLAG	**0.360**	ok
Aug 2024	−8.4%	9.1%	**0.363**	ok	0.346	ok

FLAG rule: MinDD $< -10\%$, Frac Inv>1 $< 25\%$, Mean $P < 0.35$.

Why does the third factor fail to help? The lagged change in log(VIX) is highly correlated with movements already captured by the level of log(VIX) and by the inversion ratio. In acute panic windows the level and inversion already rise sharply, so the incremental signal is small. In grind-down windows (2015, 2022) the change factor adds noise rather than a persistent stress signal. Under an identical execution layer the extra factor therefore does not translate into better out-of-sample risk metrics or clearer episode coverage. The practical implication is deliberate: a transparent two-factor rule plus a well-designed position map is sufficient; adding more transition drivers does not pay and only increases operational complexity.

5.2. Expanding-Window Out-of-Sample Performance

Table 3 presents the primary expanding-window results under the frozen soft-mapping rule ($\tau = 0.65$, $W_{\text{STRESS}} = 0.50$). The two-factor strategy is the main result. The three-factor specification produces similar, and in some metrics slightly inferior, risk statistics and is not promoted.

Table 3: Expanding-Window Out-of-Sample Performance (Core Metrics)

Specification	Mean w	Ann. Ret.	Ann. Vol.	MaxDD	Sharpe	TO
2F soft $\tau = 0.65$, $W = 0.50$ (Main)	0.769	12.31%	**13.26%**	−27.74%	**0.93**	**1.56**
3F soft $\tau = 0.65$, $W = 0.50$ (Ext.)	0.784	12.30%	13.27%	**−26.24%**	**0.93**	1.60
2F hard $\tau = 0.50$, $W = 0$ (old)	0.942	14.85%	16.98%	−33.72%	0.87	3.51
3F hard $\tau = 0.50$, $W = 0$ (old)	0.972	**15.78%**	17.39%	−33.72%	0.91	1.99
Buy & Hold	1.000	15.66%	17.81%	−33.72%	0.88	—

Expanding window: first estimation at day 1,500, re-estimated every 130 days. Net of costs and stylized tax. TO = annualized turnover.

Table 4: Tail-Risk Metrics (Same Expanding-Window OOS Sample)

Specification	Calmar	CVaR_{95}	Downside Dev.
2F soft $\tau = 0.65$, $W = 0.50$ (Main)	0.44	$-\mathbf{1.99}\%$	11.00%
3F soft $\tau = 0.65$, $W = 0.50$ (Ext.)	**0.47**	-2.02%	**10.85**%
Buy & Hold	0.46	-2.72%	14.72%

Calmar = Ann. Ret. / |MaxDD|. CVaR_{95} is the mean of daily OOS returns at or below the 5% quantile (reported in percent). Downside deviation is the annualized standard deviation of negative daily returns.

Figure 4 displays the corresponding expanding-window equity curves and drawdowns. The two-factor strategy tracks buy-and-hold in normal periods but systematically reduces exposure during stress, producing a visibly shallower maximum drawdown while preserving most of the long-term growth.

Figure 4: Expanding-window out-of-sample equity curves (upper panel) and drawdowns (lower panel). The two-factor strategy with frozen soft mapping (red) reduces maximum drawdown by approximately six percentage points relative to buy-and-hold (gray) while keeping average equity exposure near 0.77.

Figure 5 shows the realized equity weight path under the same frozen rule. The five-day minimum holding period and (S, s) band keep turnover moderate (annualized 1.56), confirming that the strategy operates at a weekly-scale execution cadence despite daily regime detection.

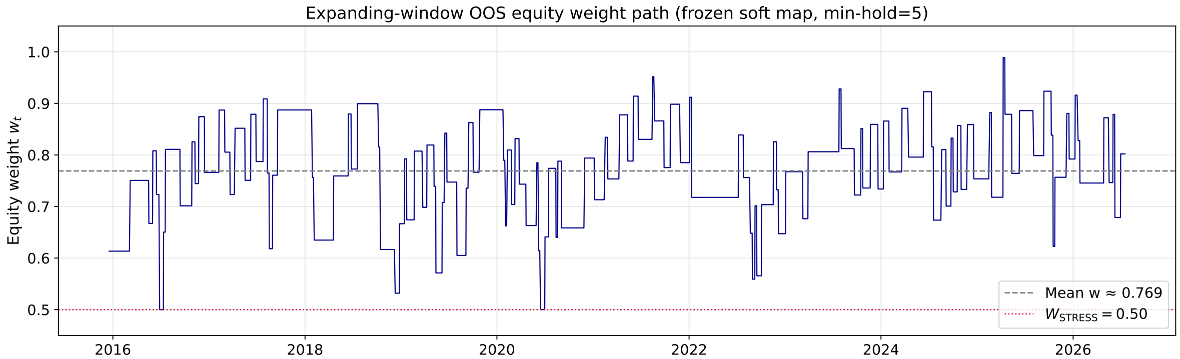

Figure 5: Out-of-sample equity weight path under the frozen soft-mapping rule ($\tau = 0.65$, $W_{\text{STRESS}} = 0.50$) and five-day minimum holding period. Average weight is 0.769; annualized turnover is 1.56.

5.3. Practitioner Benchmarks

A natural question is whether the Markov-switching rule outperforms simple heuristics that require far less computation. Table 5 places the two-factor strategy alongside three widely used alternatives: Moreira–Muir inverse-volatility scaling, a 200-day moving-average timing rule, and a naïve VIX-level threshold (reduce exposure when VIX > 25).

Table 5: Practitioner Benchmarks (Expanding-Window OOS)

Strategy	Mean w	Ann. Ret.	Ann. Vol.	MaxDD	Sharpe	Calmar
2F soft (Main)	0.769	12.31%	13.26%	−27.74%	0.93	0.44
Inverse-Vol (Moreira–Muir)	0.828	11.62%	**11.47%**	**−15.40%**	**1.01**	**0.75**
200-day MA Timing	0.830	11.77%	11.85%	−18.94%	0.99	0.62
VIX $>$ 25 Threshold	0.929	12.78%	13.79%	−21.08%	0.93	0.61
Buy & Hold	1.000	**15.66%**	17.81%	−33.72%	0.88	0.46

All figures net of costs and stylized tax. Calmar = Ann. Ret. / |MaxDD|. Inverse-Vol uses a 21-day lagged realized volatility and a 12% target volatility. MA200 and VIX>25 signals are lagged one day.

Two patterns stand out. First, all active rules reduce maximum drawdown relative to buy-and-hold; the improvement is largest for inverse-volatility scaling (MaxDD −15.4%, Calmar 0.75). Second, the two-factor regime rule sits between buy-and-hold and the simple heuristics on drawdown, while delivering the lowest turnover among the active rules (annualized 1.56 versus roughly 3–5 for the benchmarks). The regime rule is therefore not a dominant risk-adjusted alternative to inverse-volatility scaling; its practical case rests on a transparent, low-turnover mapping from a forward-looking stress probability to position size, together with explicit constraints (minimum holding period and stylized tax drag) that the pure inverse-volatility rule does not encode. While simple heuristics like inverse-volatility scaling achieve a lower maximum drawdown (-15.4%), they typically demand frequent rebalancing and higher trading intensity. The proposed 2F TVTP regime model serves as an intermediate, low-turnover alternative—achieving an annualized turnover of just

1.56 compared to roughly 3–5 for active benchmarks. It delivers meaningful drawdown reduction (from -33.7% down to -27.7%) relative to buy-and-hold while preserving long-term equity growth under strict execution constraints (a five-day minimum holding period and stylized tax drag). The primary contribution is therefore not a claim of absolute Sharpe dominance, but rather the provision of an implementable, forward-looking risk overlay that balances tail-risk mitigation against operational frictions.

5.4. Robustness and the Role of the Execution Layer

Table 6 reports the two-factor execution-layer sensitivity grid. Hard thresholding fails to improve maximum drawdown. An aggressive soft map over-hedges. The intermediate calibration $\tau = 0.65$, $W_{\mathrm{STRESS}} = 0.50$ achieves the target average weight range while delivering material risk reduction. This confirms that the execution layer is frequently the binding constraint on out-of-sample performance.

Table 6: Two-Factor Execution-Layer Sensitivity

Specification	Mean w	Ann. Ret.	Ann. Vol.	MaxDD	Sharpe	TO
hard $\tau = 0.50$, $W = 0$	0.942	**14.85%**	16.98%	−33.72%	0.87	3.51
soft $\tau = 0.45$, $W = 0$	0.363	4.11%	**5.80%**	**−13.84%**	0.71	7.39
soft $\tau = 0.55$, $W = 0.30$	0.611	8.96%	10.03%	−23.03%	0.89	3.78
soft $\tau = 0.60$, $W = 0.40$	0.689	10.57%	11.57%	−23.88%	0.91	2.70
soft $\tau = 0.65$, $W = 0.50$ (Frozen)	0.769	12.31%	13.26%	−27.74%	**0.93**	**1.56**

An expanded grid covering $\tau \in [0.50, 0.80]$, $W_{\mathrm{STRESS}} \in [0.30, 0.70]$, and minimum holding periods of 3–10 days yields the same qualitative conclusion: the frozen calibration sits in a stable region that balances risk reduction against average equity exposure. Extreme settings either restore buy-and-hold drawdowns or sacrifice too much return.

The pattern is consistent with the short unconditional Stress duration documented in Table 1. Because the filtered probability remains elevated for only a few days on average, a hard threshold combined with a minimum-holding constraint rarely triggers de-risking. The soft map solves this problem by allowing partial position reduction even when the probability is only moderately elevated, while the frozen parameters prevent over-hedging.

6. Implementation Considerations

Section 6 discusses practical implementation considerations, highlighting how the hybrid daily-detection / weekly-execution framework allows institutional risk overlays or systematic wealth managers to operate under low-turnover and transaction-drag constraints. Statistical regime detection alone is insufficient for practical use. Transaction costs, capital-gains tax considerations, and institutional aversion to excessive rebalancing impose material frictions. The soft-mapping rule converts the filtered Stress probability into a continuous equity weight. With the frozen parameters $\tau = 0.65$ and $W_{\text{STRESS}} = 0.50$, average equity exposure remains in the range 0.75–0.85. An (S, s)-style band and a five-trading-day minimum holding period further limit turnover. The net effect is that the overlay detects regimes at daily frequency yet effectively rebalances at a weekly horizon, making it suitable as a systematic downside-protection layer rather than a high-turnover alpha engine.

Transaction costs are charged proportionally to absolute turnover. A stylized tax drag is applied whenever a position reduction crystallizes a gain. Because the minimum-holding constraint already limits turnover, the incremental tax drag remains modest. Sensitivity analysis (Table 6) confirms that deviations from the frozen calibration either restore buy-and-hold drawdowns or sacrifice too much return, reinforcing that the execution layer is a first-order determinant of practical performance.

The following outline is intended for institutional risk-management or systematic wealth-management implementation:

1. Obtain daily SPY, VIX, VIX3M, and a cash proxy (e.g., BIL or a money-market fund). All signals must be lagged one trading day.

2. Update the two-factor filtered probability $P(\text{Stress})$ using the most recent parameter estimates (re-estimated roughly every six months, or approximately every 130 trading days).

3. Apply the soft map

$$w^* = 1 + (0.50 - 1) \cdot \text{clip}\big(P(\text{Stress})/0.65,\ 0,\ 1\big).$$

4. Change the actual portfolio weight only if (a) at least five trading days have elapsed since the last change and (b) the absolute difference between the target and the current weight exceeds a small band (e.g., 5–10 percentage points).

5. In tax-advantaged or institutional accounts the stylized tax drag is typically zero or negligible; the five-day minimum holding period already reduces the incidence of short-term crystallizations.

6. The entire pipeline can be maintained within a production risk system or a compact automated script; no continuous optimization is required between re-estimation dates.

Daily statistical signals are retained for timely detection, yet the position engine deliberately slows the translation of those signals into trades. This hybrid cadence is the practical counterpart of the statistical finding that Stress regimes are short-lived: without the soft map and minimum-holding constraint, the model would either trade too frequently or fail to reduce risk at all. Framed as a dynamic risk overlay, the rule prioritizes controlled drawdown mitigation and operational simplicity over the pursuit of superior risk-adjusted returns.

7. Conclusion

Large maximum drawdowns permanently impair the terminal wealth of long-horizon investors. This paper develops a hybrid **dynamic risk-overlay** framework that addresses the problem under realistic trading frictions. A two-state Markov-switching GJR-GARCH model with time-varying transition probabilities driven by the lagged log(VIX) and the VIX term-structure inversion generates daily Stress probabilities. These probabilities are mapped into continuous equity weights via a soft threshold rule, subject to a five-trading-day minimum holding period and stylized tax considerations.

Under strict expanding-window out-of-sample evaluation the overlay reduces annualized volatility by approximately **25 percent** and maximum drawdown by roughly **six percentage points** relative to buy-and-hold, while keeping turnover modest and preserving a comparable Sharpe ratio. A three-factor extension does not improve risk metrics or episode coverage and is retained only as a robustness check. The binding constraint on out-of-sample risk reduction is frequently the execution-layer mapping from probability to position size.

The VIX inversion ratio remains a useful but incomplete signal: it performs well in acute panic episodes yet systematically under-detects prolonged grind-down markets. Short unconditional Stress duration is a structural feature of the estimated dynamics; the soft-mapping rule is a pragmatic remedy rather than a statistical solution. These limitations are acknowledged rather than papered over.

The contribution is therefore practical. We provide a transparent, implementable downside-protection rule that demonstrably limits large drawdowns without claiming superior risk-adjusted returns under rigorous out-of-sample conditions. Future work may explore richer VIX-derived features, multi-asset extensions, and more granular models of crisis-period trading costs.

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